Steady-State Analysis of the Parallel Resonant Converter with LLCC-Type Commutation Network

Issa Batarseh, *Member, IEEE* and C. Q. Lee, *Member, IEEE*

***Abstract*—By using a proper transformation of state variables, a steady-state analysis for the 4th order high frequency LLCC-type parallel resonant converter (PRC-LLCC) operating in the continuous conduction mode is presented. The method makes use of the two-dimensional state-plane diagram which has been used successfully to analyze resonant converters of lower order. Based on this analysis, a set of control characteristic curves have been derived from which a comparison between this converter and the conventional PRC is presented. Our theoretical results are verified by computer simulations. A design procedure with a specific design example will be also given.**

Introduction

THE resonant concept still enjoys a high degree of acceptance in the design of smaller and lighter high frequency dc-to-dc converters. Thus far, the conventional parallel and series resonant dc-to-dc converters, where the tank circuits consist of a single capacitor and inductor, have been thoroughly analyzed in the literature. Recently, interest in resonant converters is shifting toward high order resonant converters in which the tank circuit has more than two reactive components [1]–[7]. The advantages of studying high order PRC are: 1) At high enough frequencies, the parasitic circuit inductances and junction capacitors of transistors and diodes cannot be ignored and their inclusion in the converter design is unavoidable. Hence, practical converters are inherently high order. 2) It has been shown before that the addition of reactive components to the tank circuit has resulted in a substantial improvement in the performance characteristics of the resonant converters [1]–[5]. 3) High order converters allow the generation of new converter topologies. As a result, a designer has the choice of using the optimum or suitable configuration for particular applications. Results from the analysis of the 3rd order LCC- and LLC-type parallel resonant converters show that the addition of a capacitor and inductor to the conventional LC-type PRC, respectively, has resulted in many desirable characteristics. For example, at switching frequencies lower than the resonant frequency, the voltage gain of the LCC-type converter is lower than the gain of the conventional PRC which allows a lower turn-ratio transformer, and higher frequency of operation in the converter design. Moreover, its control characteristic curves reveal that for some range of γ, the converter gain becomes virtually independent of the load [1]. In the case of the LLC-type converter, in addition to having good control characteristics, it has a voltage gain even lower than that of LCC-type converter [7].

In this paper, we present a new converter topology in which the tank circuit consists of two inductors and two capacitors. As far as the performance characteristics is concerned, this LLCC-type converter topology combines the advantages of both the LCC-type and the LLC-type converters. At present, there is little theoretical method available to analyze such converters [8]. We present a relatively simple state-plane diagram technique to derive the steady state response of the converter, from which the performance characteristics can be derived. Computer simulation results for a given specification are also presented.

Manuscript received May 30, 1989; revised May 31, 1990. This paper was presented at the 1989 IEEE Power Electronics Specialists Conference, Milwaukee, WI, June 26-29.

C. Q. Lee is with the Department of Electrical Engineering and Computer Science, University of Illinois at Chicago, P.O. Box 4348, Chicago, IL 60690.

I. Batarseh is with the Department of Electrical Engineering, University of Central Florida, Orlando, FL 32816.

IEEE Log Number 9100530.

Circuit Description

The idealized half-bridge LLCC-type parallel resonant converter using bidirectional switches is shown in Fig. 1. The commutational network of this topology consist of two resonant tanks: the L_s–C_s series tank and the L_p–C_p parallel tank. Unlike previously presented resonant converters, this circuit represents a 4th order nonlinear system whose steady state dynamic properties have not been studied before. In this section, the steady state operation of this converter when operated in the continuous conduction mode is presented.

To simplify the analysis, we assume that the diode-transistor circuits D_1/T_1 and D_2/T_2 are ideal and can be represented by a pair of bidirectional switches operated at 50% duty cycle over a switching period T_s. Hence, we may represent the effect of the switching devices by means of an equivalent square wave voltage source $v_s(t)$ with amplitude equal $\pm V_g$. Furthermore, since a large value of the low pass filter inductor L_o is used at the output circuit, the load current I_o is considered constant. Consequently, the rectifying circuit is equivalent to a dependent current

0885-8993/91/0700-0525$01.00 © 1991 IEEE

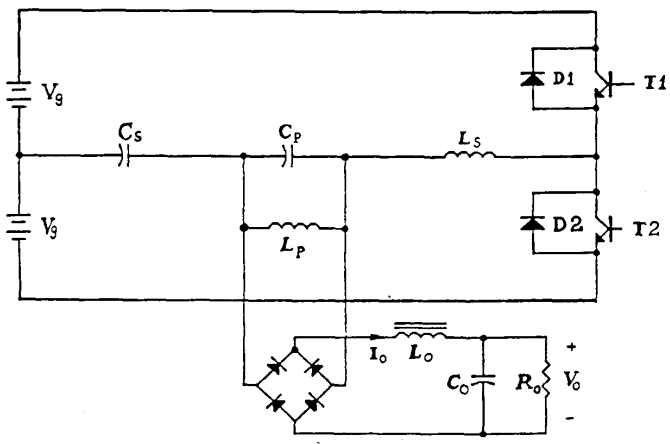


Fig. 1. Half-bridge LLCC-type parallel resonant converter circuit.

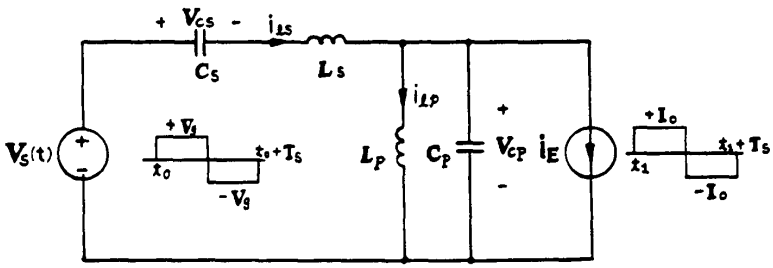


Fig. 2. Equivalent circuit model of Fig. 1.

sink $i_E(t)$ with amplitudes equal to $+I_o$ and $-I_o$, depending on whether the voltage $v_{cp}(t)$ is positive or negative, respectively. The simplified equivalent circuit for the LLCC-type PRC of Fig. 1 is shown in Fig. 2.

The basic circuit operation of this new topology and the LC-type PRC are similar. Typical steady-state voltage and current waveforms for the LLCC-type PRC operating in the continuous conduction mode are shown in Fig. 3. The converter operation can be described as follows: let t_o to be the time at which the converter's operation starts by turning on T_1 and the capacitor voltages and inductor currents at this time are given by $v_{cp}(t_o)$, $v_{cs}(t_o)$, $i_{Ls}(t_o)$ and $i_{Lp}(t_o)$, respectively. At this time, we assume $v_{cp} < 0$. At $t = t_1$, the capacitor voltage v_{cp} changes sign and the current i_E reverses direction. At $t = t_2$, the inductor current $i_{Ls}(t)$ crosses zero and transistor T_1 is naturally turned off and flyback diode D_1 starts to conduct in order to maintain current continuity in the resonant inductor L_s. To allow for a continuous conduction mode of operation, the current which flows in D_1 is forced to turn off at $t = t_o + T_s/2$, when T_2 is turned on. Similarly, when the inductor current $i_{Ls}(t)$ passes zero, T_2 is naturally turned off and D_2 starts to carry the inductor L_s current in the opposite direction. Again at $t = t_3$, the capacitor voltage of C_p changes polarity and the current $i_E(t)$ inverses direction. At $t = t_o + T_s$, the cyclic operation is repeated. Similar to the conventional and the LCC-type converters, when operated in the continuous conduction mode, this converter has four possible circuit modes of operation, namely; $M1$, $M2$, $M3$ and $M4$ as shown in Fig. 3. Fig. 4 shows the corresponding four circuit topologies. Table I illustrates these modes and the conditions under which they occur.

Steady-State Analysis

A. Mathematical Description

The steady-state response may be derived from the idealized equivalent circuit of Fig. 2. The voltages v_s, v_{cs}, v_{cp} and currents i_{Ls}, i_{Lp}, and i_E are defined according to this figure. As stated above, the rectifying circuit can be replaced by a dependent current source $i_E(t)$ with current polarity depending on the sign of $v_{cp}(t)$ such that the average power dissipated in it is equal to the dc output power of the converter. We may represent $i_E(t)$ by

$$i_E(t) = \frac{v_{cp}(t)}{|v_{cp}(t)|} I_o \tag{1}$$

where $|v_{cp}(t)|$ is the magnitude of $v_{cp}(t)$ and I_o is equal to V_o/R_o where

$$V_o = \frac{1}{T_s} \int_o^{T_s} |v_{cp}(t)|\, dt \tag{2}$$

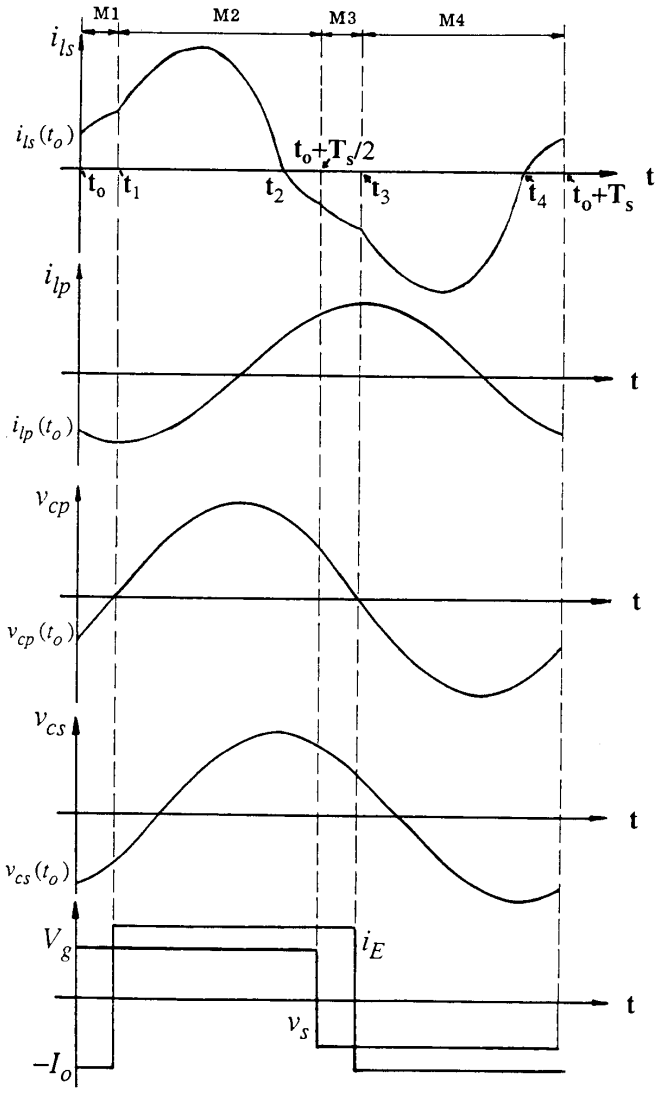


Fig. 3. Typical waveforms for i_{Ls}, i_{Lp}, v_{cp}, v_{cs}, v_s, and i_E of the PRC-LLCC.

and T_s is the switching period. The following differential equations give the mathematical description for the circuit model of Fig. 2:

$$\frac{dv_{cp}(t)}{dt} = \frac{1}{C_p}\,[i_{Ls}(t) - i_{Lp}(t) - i_E(t)] \tag{3}$$

$$\frac{dv_{cs}(t)}{dt} = \frac{1}{C_s}\,i_{Ls}(t) \tag{4}$$

$$\frac{di_{Ls}(t)}{dt} = \frac{1}{Ls}\,[v_s(t) - v_{cs}(t) - v_{cp}(t)] \tag{5}$$

$$\frac{di_{Lp}(t)}{dt} = \frac{1}{L_p}\,v_{cp}(t). \tag{6}$$

Equations (3)–(6) represent a fourth order system with a nonlinear dependent current source which are difficult to solve by the conventional circuit approach. However, by using a proper transformation on the state variables v_{cs}, v_{cp}, i_{Ls}, and i_{Lp}, it will be shown that the steady state solution for (3)–(6) can be derived from two-dimensional state-plane diagrams.

B. Derivation of State-Plane Equations

It has been shown that the state-plane design approach is a successful method to analyze 2nd and 3rd order converters [1], [2]. Unlike the state variable transformation used for these converters, the state variable transformation for the 4th order converter is more complex. It is

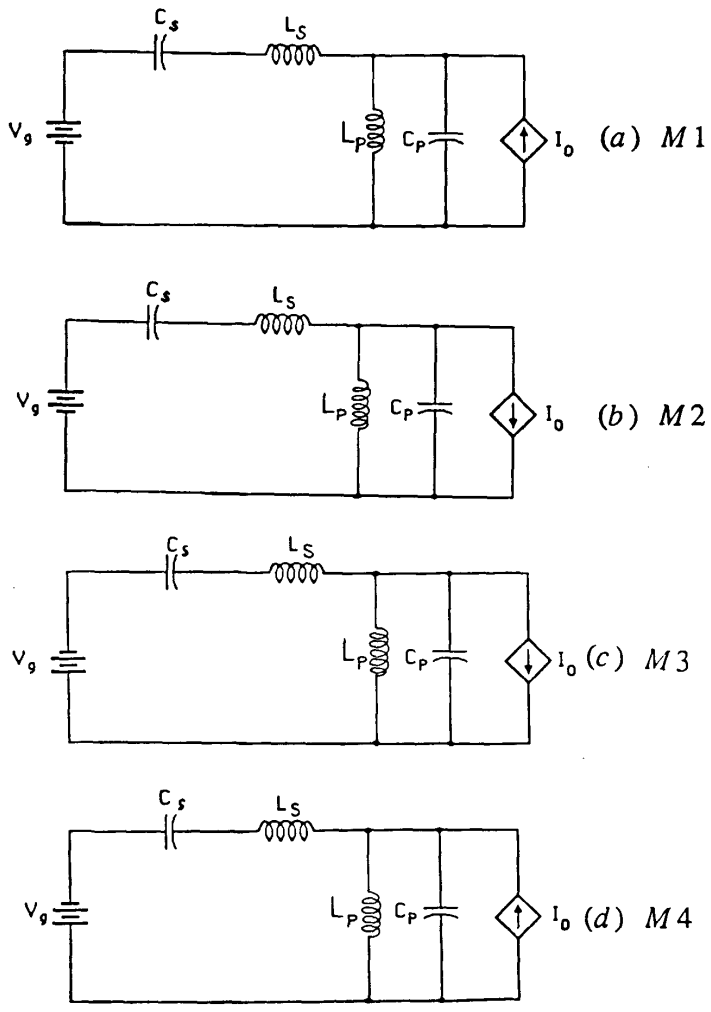


Fig. 4. Four topological circuit modes of Fig. 2.

TABLE I
MODES OF OPERATION FOR LLCC-TYPE PRC

Mode	Switches On	Sign of the Rectified Voltage	$v_s(t)$	i_E
M_1	T_1	$v_{cp} < 0$	$+V_g$	$-I_0$
M_2	T_1/D_1	$v_{cp} > 0$	$+V_g$	$+I_0$
M_3	T_2	$v_{cp} > 0$	$-V_g$	$+I_0$
M_4	T_1/D_2	$v_{cp} < 0$	$-V_g$	$-I_0$

clear from (3)–(6) that the circuit variables v_{cs}, v_{cp}, i_{Ls} and i_{Lp} are all coupled and their steady state representation is only possible in a four dimensional state-plane. However, it can be shown that there exists a new set of state variables from which the steady-state response of the converter can be obtained from a pair of two-dimensional state planes.

Let us define the following state variable transformation where $v_1(t)$, $v_2(t)$, $i_1(t)$, and $i_2(t)$ are the new state variables,

$$\begin{bmatrix} v_1(t) \\ v_2(t) \\ i_1(t) \\ i_2(t) \end{bmatrix} = \begin{bmatrix} 1 & K_1 & 0 & 0 \\ 1 & K_2 & 0 & 0 \\ 0 & 0 & 1 & K_3 \\ 0 & 0 & 1 & K_4 \end{bmatrix} \begin{bmatrix} v_{cp}(t) \\ v_{cs}(t) \\ i_{Ls}(t) \\ i_{Lp}(t) \end{bmatrix}. \quad (7)$$

The objective of using this transformation is to find values for K_1, K_2, K_3, and K_4 such that the state variable pairs (v_1, i_1) and (v_2, i_2) are decoupled from each other so that their steady state responses can be represented by closed contours in the $v_1 - i_1$ and $v_2 - i_2$ state-planes. We should note that the choice of this transformation is not an arbitrary one. However, due to space limitations, the derivation of this transformation is not given. Future papers will carry the detailed derivation for this converter and others.

It can be shown that by substituting (7) into (3)–(6), the following state-plane equations can be obtained:

$$\frac{dv_1(t)}{di_1(t)} = \frac{L_s}{C_s}\left(1 + \frac{C_s}{C_p}K_1\right)\left[\frac{i_1(t) + K_3 i_E(t)}{v_s(t) - v_1(t)}\right] \quad (8)$$

$$\frac{dv_2(t)}{di_2(t)} = \frac{L_s}{C_s}\left(1 + \frac{C_s}{C_p}K_2\right)\left[\frac{i_2(t) + K_4 i_E(t)}{v_s(t) - v_2(t)}\right] \quad (9)$$

where

$$K_1 = 1 - \frac{L_s}{L_p}K_3 \quad (10)$$

$$K_2 = 1 - \frac{L_s}{L_p}K_4 \quad (11)$$

$$K_3 = \frac{-K_1}{\frac{C_p}{C_s} + K_1} \quad (12)$$

$$K_4 = \frac{-K_2}{\frac{C_p}{C_s} + K_2}. \quad (13)$$

From these equations, K_1, K_2, K_3, and K_4 can be solved in terms of C_p/C_s and L_s/L_p as shown next:

$$K_1 = \frac{1}{2}\left[1 + \frac{L_s}{L_p} - \frac{C_p}{C_s} + \sqrt{\left(1 + \frac{L_s}{L_p} - \frac{C_p}{C_s}\right)^2 + 4\frac{C_p}{C_s}}\right] \quad (14)$$

$$K_2 = \frac{1}{2}\left[1 + \frac{L_s}{L_p} - \frac{C_p}{C_s} - \sqrt{\left(1 + \frac{L_s}{L_p} - \frac{C_p}{C_s}\right)^2 + 4\frac{C_p}{C_s}}\right] \quad (15)$$

$$K_3 = \frac{1}{2}\left[1 + \frac{L_p}{L_s}\frac{C_p}{C_s} - \frac{L_p}{L_s} + \sqrt{\left(1 - \frac{L_p}{L_s}\frac{C_p}{C_s} + \frac{L_p}{L_s}\right)^2 + 4\frac{C_p}{C_s}\left(\frac{L_p}{L_s}\right)^2}\right] \quad (16)$$

$$K_4 = \frac{1}{2}\left[1 + \frac{L_p}{L_s}\frac{C_p}{C_s} - \frac{L_p}{L_s} - \sqrt{\left(1 - \frac{L_p}{L_s}C_s\frac{C_p}{C_s} + \frac{L_p}{L_s}\right)^2 + 4\frac{C_p}{C_s}\left(\frac{L_p}{L_s}\right)^2}\right]. \quad (17)$$

If we normalize $v_1(t)$ and $v_2(t)$ by V_g, $i_1(t)$ and $i_2(t)$ by V_g/Z_{01} and V_g/Z_{02}, respectively, we obtain from (8) and (9) that,

$$\frac{dv_{n1}(t)}{di_{n1}(t)} = \frac{i_{n1}(t) + K_3 i_{nE1}}{v_{ns}(t) - v_{n1}(t)} \quad (18)$$

$$\frac{dv_{n2}(t)}{di_{n2}(t)} = \frac{i_{n2}(t) + K_4 i_{nE2}}{v_{ns}(t) - v_{n2}(t)} \quad (19)$$

where

$$Z_{01} = \sqrt{\frac{L_s}{C_s}\left(1 + \frac{C_s}{C_p}K_1\right)} \quad (20)$$

$$Z_{02} = \sqrt{\frac{L_s}{C_s}\left(1 + \frac{C_s}{C_p}K_2\right)} \quad (21)$$

$$i_{nE1} = \frac{Z_{01} i_E}{V_g} \quad (22)$$

$$i_{nE2} = \frac{Z_{02} i_E}{V_g}. \quad (23)$$

Equations (18) and (19) can be solved by a piece wise time invariant analysis, similar to that of the conventional PRC. Over a switching period, it can be shown that there exist four time intervals where the solutions consists of circular arcs in the $v_{n1} - i_{n1}$ and $v_{n2} - i_{n2}$ state-planes and they are given by

$$[v_{n1}(t) - v_{nsi}(t)]^2 + [i_{n1}(t) - K_3 i_{nE1i}(t)]^2 = |V_{1ni}|^2 \quad (24)$$

$$[v_{n2}(t) - v_{nsi}(t)]^2 + [i_{n2}(t) - K_4 i_{nE2i}(t)]^2 = |V_{2ni}|^2 \quad (25)$$

where $i = 1, 2, 3, 4$. V_{1ni} and V_{2ni} are the radii.

The centers of these circular arcs correspond to the singular points of (18) and (19). It can be shown that these singular points have the following values in respective state planes.

$v_{n1} - i_{n1}$ plane	$v_{n2} - i_{n2}$ plane
(v_{ns1}, i_{nE11}): $(1, -K_3 I_{n01})$	(v_{ns1}, i_{nE21}): $(1, -K_4 I_{n02})$
(v_{ns2}, i_{nE12}): $(1, K_3 I_{n01})$	(v_{ns2}, i_{nE22}): $(1, K_4 I_{n02})$
(v_{ns3}, i_{nE13}): $(-1, K_3 I_{n01})$	(v_{ns3}, i_{nE23}): $(-1, K_4 I_{n02})$
(v_{ns4}, i_{nE14}): $(-1, -K_3 I_{n01})$	(v_{ns4}, i_{nE24}): $(-1, -K_4 I_{n02})$

(26)

where

$$I_{n01} = \frac{I_0 Z_{01}}{V_g} \quad (27)$$

$$I_{n02} = \frac{I_0 Z_{02}}{V_g}. \quad (28)$$

C. Steady-State Trajectories

Based on the above discussion, steady state trajectories for this converter could be constructed in the $v_{n1} - i_{n1}$ and $v_{n2} - i_{n2}$ state planes. It can be shown that typical state-plane diagrams in the $v_{n1} - i_{n1}$ and $v_{n2} - i_{n2}$ planes over a switching period are shown in Fig. 5(a) and (b), respectively. The portions shown by the solid and the broken lines represent the steady-state responses over the half switching cycle when $v_{ns} = 1$ and $v_{ns} = -1$, respectively.

The sum $\beta_1 + \beta_2 + \alpha$ and $\beta_1' + \beta_2' + \alpha'$ represent the frequency parameters γ_1 and γ_2, respectively, and they are defined as the ratio of the resonant to switching frequencies, i.e.,

$$\gamma_1 = \beta_1 + \beta_2 + \alpha = \omega_{01}\frac{T_s}{2} = \pi\frac{f_{01}}{f_s} \tag{29}$$

$$\gamma_2 = \beta_1' + \beta_2' + \alpha' = \omega_{02}\frac{T_s}{2} = \pi\frac{f_{02}}{f_s} \tag{30}$$

where

$$\omega_{01}^2 = \frac{1}{L_s C_s}\left(1 + \frac{C_s}{C_p}K_1\right) = \frac{Z_{01}^2}{L_s^2} \tag{31}$$

$$\omega_{02}^2 = \frac{1}{L_s C_s}\left(1 + \frac{C_s}{C_p}K_2\right) = \frac{Z_{02}^2}{L_s^2}. \tag{32}$$

It can be shown that ω_{o1} and ω_{o2} are the resonant frequencies at which the input impedance looking into the tank circuit from the source terminal with the load open circuited is zero. Based on the definition of I_{n01} and I_{n02} given (27) and (28), we can derive the following relation from (31) and (32) that,

$$\frac{I_{n01}}{I_{n02}} = \frac{K_3}{K_4}\sqrt{\frac{1 + C_s/C_pK_1}{1 + C_s/C_pK_2}}. \tag{33}$$

From (29)–(33), it is easy to show that the following relations hold:

$$\frac{\gamma_1}{\gamma_2} = \frac{Z_{01}}{Z_{02}} = \frac{\omega_{01}}{\omega_{02}} = \frac{K_4 I_{n01}}{K_3 I_{n02}} = \sqrt{\frac{1 + C_s/C_pK_1}{1 + C_s/C_pK_2}}. \tag{34}$$

It is important to note that once C_p/C_s and L_s/L_p are known, then (34) gives a direct relation between the parameters of the two state planes.

Design Parameters

The converter dc output voltage can be calculated from

$$V_0 = \frac{1}{T_s/2}\left[\int_{t_0}^{t_1} - v_{cp}(t)\,dt + \int_{t_1}^{t_0+T_s/2} v_{cp}(t)\,dt\right]. \tag{35}$$

From (7), $v_{cp}(t)$ is given by

$$v_{cp}(t) = \frac{v_1(t) - v_2(t)}{K_1 - K_2}. \tag{36}$$

Substituting (36) into eq. (35) and using (8), we obtain the following normalized output voltage which is also the voltage gain of the converter:

$$\frac{V_0}{V_g} = M = \frac{2}{K_1 - K_2}\left[\frac{i_{n1}(t_1)}{\gamma_1} - \frac{i_{n2}(t_1)}{\gamma_2}\right]. \tag{37}$$

The average input current is given by

$$I_g = \frac{1}{T_s/2}\left[\int_{t_0}^{t_1} i_{Ls}(t)\,dt + \int_{t_1}^{t_0+T_s/2} i_{Ls}(t)\,dt\right]. \tag{38}$$

From (7), $i_{Ls}(t)$ may be written as

$$i_{Ls}(t) = \frac{K_4 i_1(t) - K_3 i_2(t)}{K_4 - K_3}. \tag{39}$$

Substituting (39) into (38) and using (9), we obtain the following normalized average input current:

$$I_{ng} = \frac{-2}{\gamma_1}\left[\frac{K_4 v_{n1}(t_o)}{(K_4 - K_3)} + \frac{(1 + K_4)K_3 v_{n2}(t_o)}{(K_4 - K_3)(1 + K_3)}\right]. \tag{40}$$

Since we assume lossless resonant circuit, conservation of power requires that $I_g V_g = I_o V_o$. Applying this relation to (37) and (40), and after some algebraic manipulation, the following relations are obtained:

$$i_{n1}(t_1) = \frac{-v_{n1}(t_o)}{|K_3 I_{n01}|} \tag{41}$$

$$i_{n2}(t_1) = \frac{+v_{n2}(t_o)}{|K_4 I_{n02}|}. \tag{42}$$

It can be shown that these relations can also be obtained directly from the state diagrams given in Fig. 5(a) and (b), respectively.

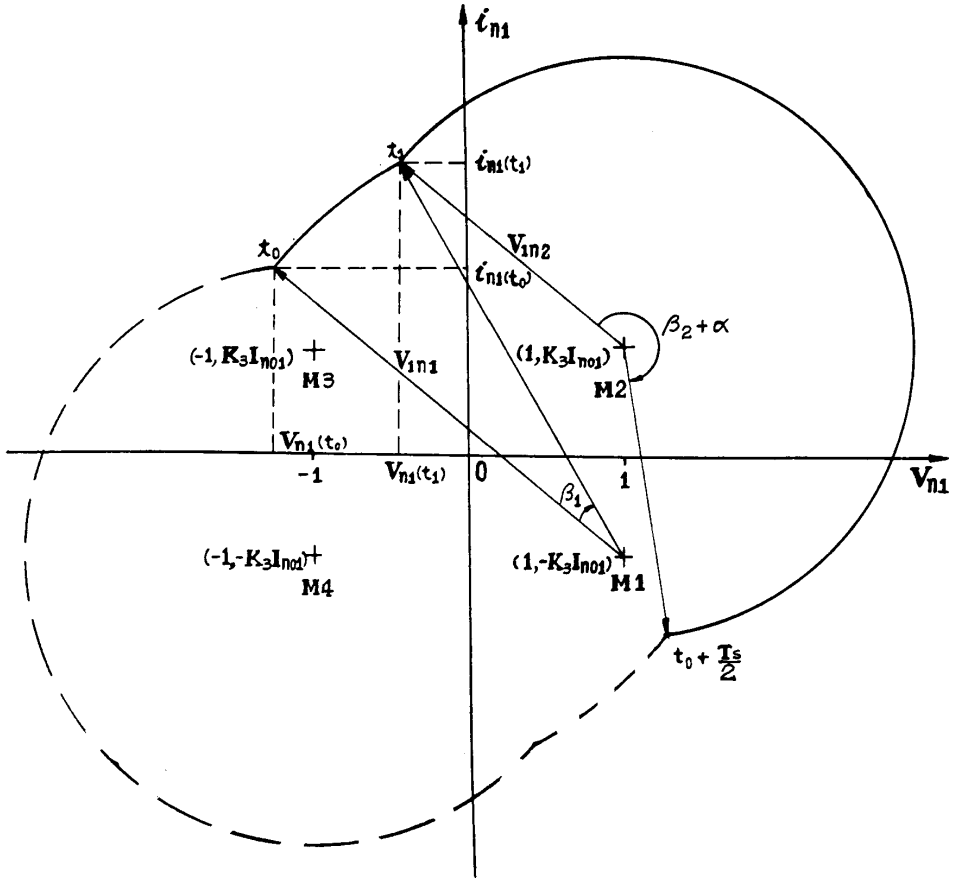
Characteristic Parameters

It can be shown that for given C_p/C_s, L_s/L_p and either I_{n01} or I_{n02}, the steady-state responses can be represented by a family of closed trajectories in the $v_{n1} - i_{n1}$ and $v_{n2} - i_{n2}$ state planes. A unique state-plane trajectory is determined if either γ_1 or γ_2 is also specified. The relations between the initial switching voltage and current given below are obtained directly from state-plane diagrams of Fig. 5(a) and (b), respectively.

$$i_{n1}(t_o) = \pm\sqrt{\left[\frac{1}{(K_3 I_{n01})^2} - 1\right]v_{n1}^2(t_0) + (K_3 I_{n01})^2 - 1 + (v_{n1}(t_1) - 1)^2} - |K_3 I_{n01}| \tag{43}$$

$$i_{n2}(t_o) = \pm\sqrt{\left[\frac{1}{(K_4 I_{n02})^2} - 1\right]v_{n2}^2(t_0) + (K_4 I_{n02})^2 - 1 + (v_{n2}(t_1) - 1)^2} - |K_4 I_{n02}|. \tag{44}$$

Since $v_{cp}(t) = 0$ when $t = t_1$, it can be seen from eq. (36) that $v_{n1}(t) = v_{n2}(t)$. From (24), (25), (41), and (42), we can express $v_{n1}(t_1)$ and $v_{n2}(t_1)$ as functions of $v_{n1}(t_o)$ and $i_{n1}(t_o)$, or $v_{n2}(t_o)$ and $i_{n2}(t_o)$. Thus (43) and (44) describe the initial switching point trajectories in the state planes for a given set of singular points. The positive and negative roots in (43) and (44) correspond to the natural and



(a)

Fig. 5. Steady-state trajectories for PRC-LLCC. (a) For $v_{n1} - i_{n1}$ state-plane. (b) For $v_{n2} - i_{n2}$ state-plane.

forced commutation modes of the operation, respectively, depending on whether T_1 or D_1 conducts first at $t = t_o$. It can be shown that, when the switching frequency f_s is greater than the resonant frequency f_o, the converter is operating in the forced mode. Whereas, in the natural mode operation, the case is reversed. From the state plane diagram given in Fig. 5(a), it can be shown that

$$\beta_1 = \tan^{-1}\frac{i_{n1}(t_1) + K_3I_{n01}}{1 - v_{n1}(t_1)} - \tan^{-1}\frac{i_{n1}(t_o) + K_3I_{n01}}{1 - v_{n1}(t_o)} \tag{45}$$

$$\beta_2 + \alpha = \pi - \tan^{-1}\frac{i_{n1}(t_1) - K_3I_{n01}}{1 - v_{n1}(t_1)} - \tan^{-1}\frac{v_{n1}(t_o) + 1}{i_{n1}(t_o) + K_3I_{n01}} \tag{46}$$

where

$$\beta_1 = \omega_{01}(t_1 - t_0), \qquad \alpha + \beta_2 = \omega_{01}[(T_s/2 + t_0) - t_1] \tag{47}$$

Similarly, from Fig. 5(b), we obtain

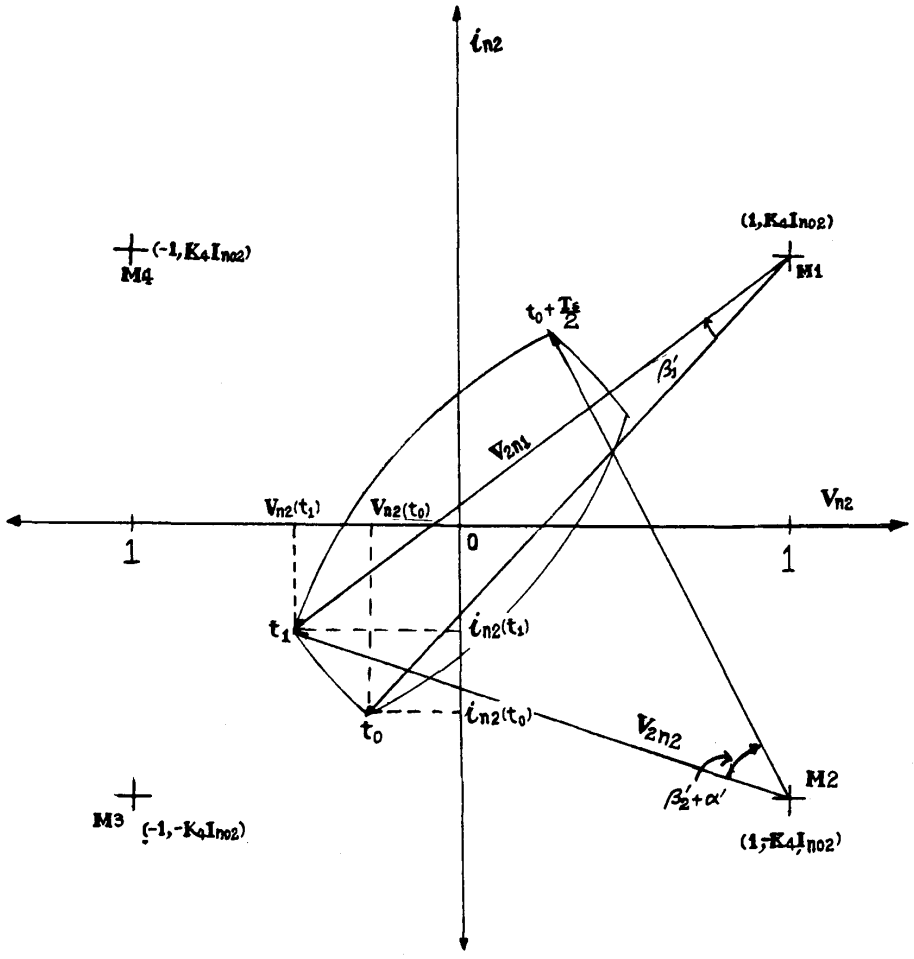
$$\beta_1' = \tan^{-1}\frac{1 - v_{n2}(t_1)}{K_4I_{n02} - i_{n2}(t_1)} - \tan^{-1}\frac{1 - v_{n2}(t_o)}{K_4I_{n02} - i_{n2}(t_o)} \tag{48}$$

$$\beta_2' + \alpha' = \tan^{-1}\frac{i_{n2}(t_o) + K_4I_{n02}}{1 + v_{n2}(t_o)} - \tan^{-1}\frac{K_4I_{n02} + i_{n2}(t_1)}{1 - v_{n2}(t_1)} \tag{49}$$

where

$$\beta_1' = \omega_{02}(t_1 - t_0), \qquad \beta_2' + \alpha' = \omega_{02}[(T_s/2 + t_0) - t_1]. \tag{50}$$

Substituting (45), (46), (48) and (49) into (29) and (30), we can derive expressions for γ_1 and γ_2 in terms of I_{n01}, I_{n02}, and the initial switching points. Furthermore, from (37), (41), and (42), the converter gain M can also be expressed in terms of the initial switching points. Hence, characteristic curves for M versus γ_1 and M versus γ_2 can be plotted under constant I_{n01} and I_{n02}. Such curves can be



(b)

Fig. 5. (*Continued*)

used to facilitate the converter design. From the design standpoint, the parameters L_s/L_p, C_p/C_s, I_{n01}, I_{n02}, M, γ_1, γ_2, Q_{p1} and Q_{p2} are of interest. Given any four of these parameters, the design of the LLCC-type converter may be obtained.

Let us define

$$Q_{p1} = \frac{R_0}{Z_{01}} \tag{51}$$

$$Q_{p2} = \frac{R_0}{Z_{02}} \tag{52}$$

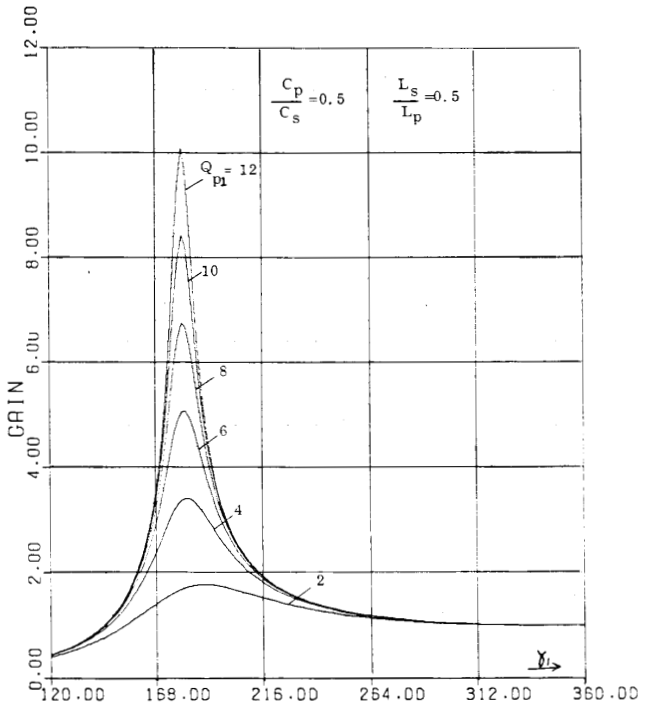
From these equations, the following relation can be derived:

$$\frac{Q_{p1}}{Q_{p2}} = \frac{Z_{02}}{Z_{01}} = \sqrt{\frac{1 + C_s/C_p K_2}{1 + C_s/C_p K_1}}. \tag{53}$$

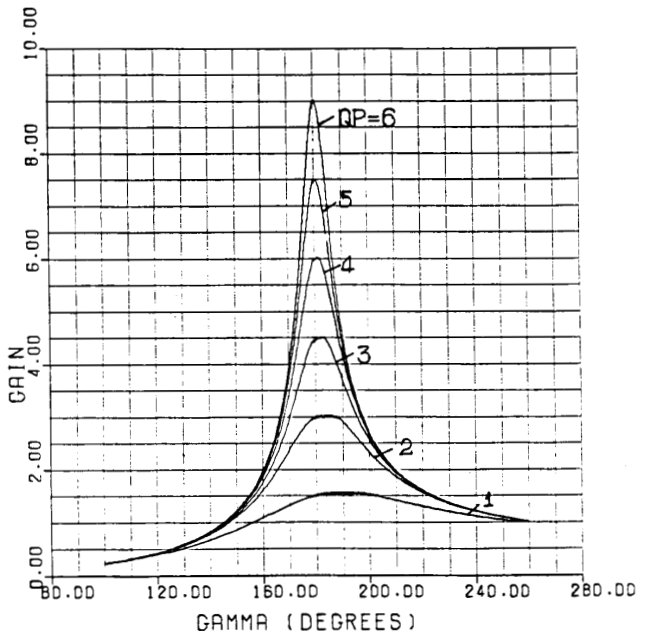
From (47) and (50), we obtain

$$\frac{\beta_1}{\beta_1'} = \frac{\beta_2 + \alpha}{\beta_2' + \alpha'} = \sqrt{\frac{1 + C_s/C_p K_1}{1 + C_s/C_p K_2}}. \tag{54}$$

Based on the above equations, it can be shown that for given C_p/C_s and L_s/L_p and any two parameters among M, γ_1, γ_2, I_{n01}, and I_{n02}, the complete converter design can be obtained. Using Q_{p1} as a constant parameter, Fig. 6(a) shows the relation between M and γ_1 for $C_p/C_s = 0.5$ and $L_s/L_p = 0.5$. The characteristic curves for the LCC-type with $C_p/C_s = 0.5$ and $L_p = \infty$ is shown in Fig. 6(b). Fig. 6(c) shows the curves for the LLC-type converter for $L_s/L_p = 0.5$ and $C_g = \infty$. Comparing these three figures, it is clear that the LLCC-type has narrower constant Q_p curves, hence, smaller γ variation is needed for output regulations.

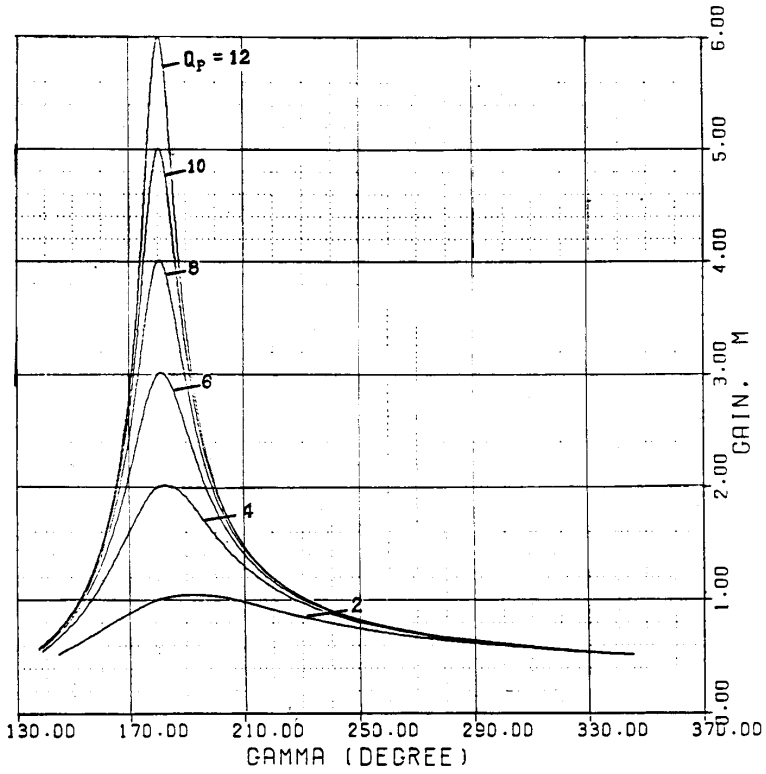


(a)



(b)

Fig. 6. Control characteristics M versus γ_1 under constant Q_{p1}: (a) LLCC-type: [C_p/C_s = 0.5 and L_s/L_p = 0.5]. (b) LCC-type: [C_p/C_s = 0.5 and L_s/L_p = ∞]. (c) LLC-type: [L_s/L_p = 0.5 and C_p/C_s = ∞].



(c)

Fig. 6. (*Continued*)

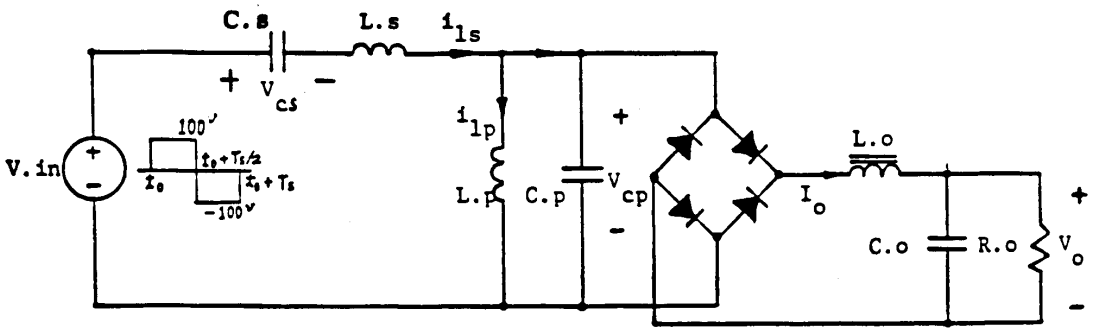


Fig. 7. LLCC-type converter circuit used for simulation.

Computer Simulation

To verify our theoretical results, a computer simulation example was performed using the following specifications:

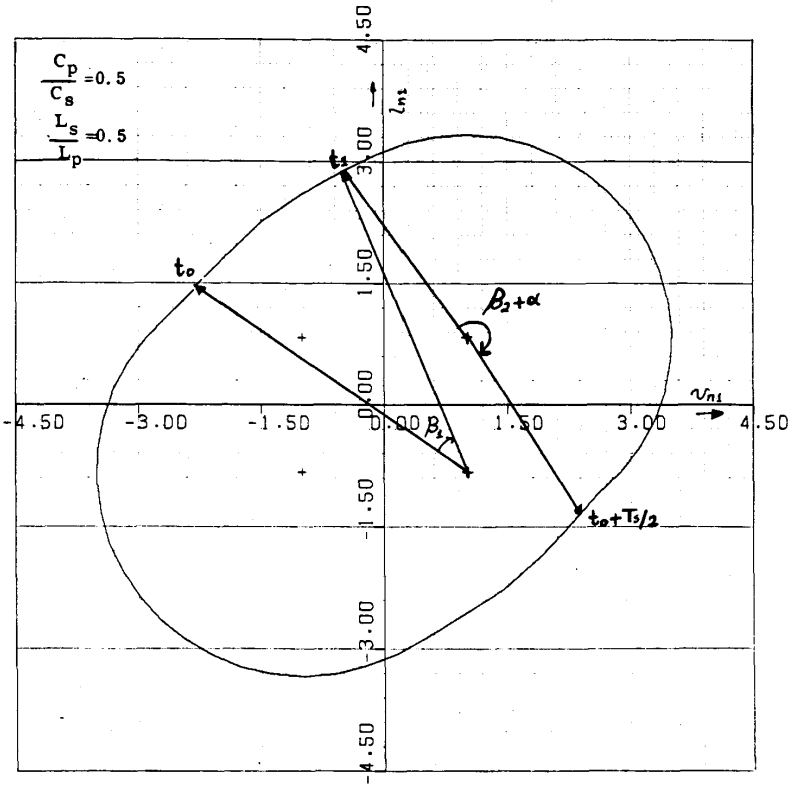
dc input voltage	$E = 2V_g = 40$ V
Switching frequency	$f_s = 500$ kHz
Output voltage	$V_o = 25$ V
Resonant frequency	$f_o = 610$ kHz

With $C_p/C_s = 0.5$ and $L_p/L_s = 0.5$, $M = 1.25$ and $\gamma_1 = 220°$, from Fig. 6(a), we obtain $Q_{p1} = 2.0$. Based on these values, the following component values are determined:

$$C_s = 318.309\ \mu\text{F}, \quad C_p = 159.154\ \mu\text{F},$$

$$L_p = 159.154\ \text{mH}, \quad L_s = 7.95722\ \text{mH}.$$

The circuit used for the simulation is shown in Fig. 7, where the circuit components used are based on the above design specifications. The tank circuit of Fig. 7 is driven by a squarewave voltage source with amplitudes equal ± 20 V and switching period $T_s = 2\ \mu$s. Simulated steady state trajectories in the $v_{n1} - i_{n1}$ and $v_{n2} - i_{n2}$ state-planes are shown in Fig. 8(a) and (b), respectively. Based on the



(a)

Fig. 8. Computer simulated state plane diagrams. (a) For $v_{n1} - i_{n1}$. (b) For $v_{n2} - i_{n2}$.

switching points of these trajectories, the theoretical values of characteristic parameters were calculated. Table II gives the comparison between the theoretical and simulated values. It is clear from the table that the results obtained from simulations are in a good agreement with those theoretically predicted. The simulated current and voltage waveforms are given in Fig. 9.

Converter Design

In this section, based on the above design characteristic curves, we give a design procedure along with a specific design example for a half-bridge PRC-LLCC converter.

The control characteristic curves for the converter are derived in terms of the design parameters: M, C_p/C_s, L_s/L_p, γ_1 and Q_{p1}. Normally, it is recommended that γ and M are specified so that the region of operation in the control characteristics under any load conditions can be determined. To outline a design procedure for the LLCC-type PRC, consider the following specifications for a 150-W converter.

minimum input voltage, $2V_{g,\min} = 230$ V.

maximum load current, $I_{o,\max} = 30$ A.

maximum switching frequency, $f_{s,\max} = 130$ kHz.

output voltage, $V_o = 5$ V.

From the design stand-point, the minimum dc supply voltage and maximum load current are given to represent the worst cases in the converter design. The following design procedure is used to determine the transformer ratio and component values:

1) Pick a proper values for γ_1 and M in Fig. 6 such that the converter operation region is away from the $\gamma_1 = 180°$ and for a smaller gain value. The choice of a smaller M is necessary for applications that require a small transformer ratio. Assume that the following design parameters are chosen:

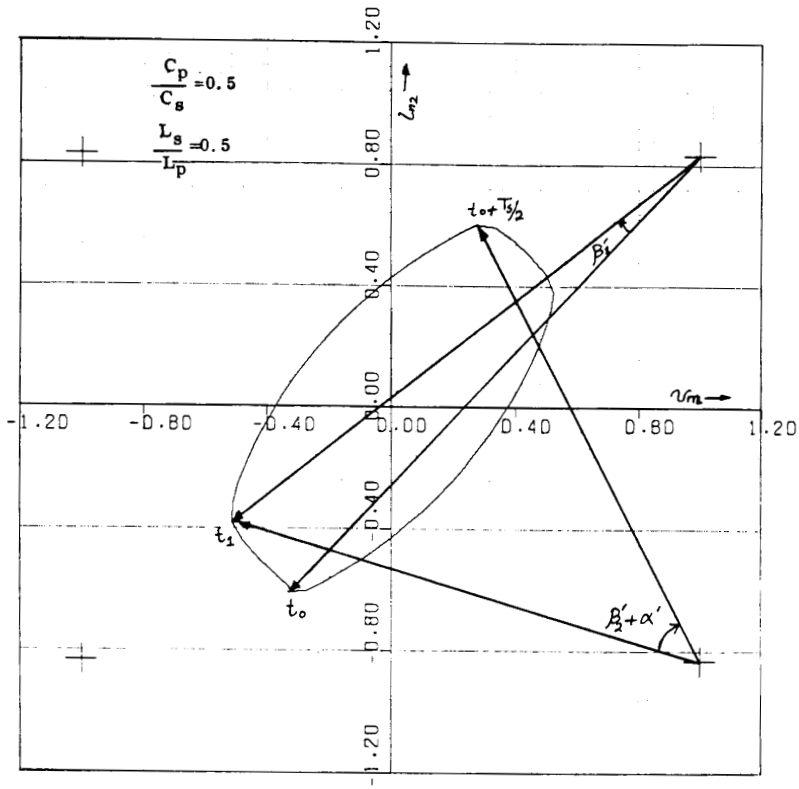
$$C_p/C_s = 0.5, \quad L_s/L_p = 0.5, \quad M = 1.25, \quad \gamma_1 = 220°.$$

2) From Fig. 6(a) we obtain $Q_{p1} = 2.0$.

3) The transformer ratio is determined from the following equation:

$$n = \frac{V_{g,\min}}{V_o} M = 23 \tag{55}$$

4) The normalized load current I_{no1} is determined from the following equation:



(b)

Fig. 8. (*Continued*)

TABLE II

State Parameter	Theoretical Value	Simulated Value
β_1	32.9°	33°
β_1'	8.42°	8.7°
$\alpha + \beta_2$	187.5°	187°
$\alpha' + \beta_2'$	47.8°	46.2°
M	1.28	1.42
γ_1	220.4°	220°
γ_2	56.22°	54.9°

$$I_{no1} = \frac{M}{Q_{p1}} = 0.625. \tag{56}$$

5) Based on the definition of the normalized output current, the following equation for the characteristic impedance Z_{o1} is obtained:

$$Z_{o1} = I_{no1} \frac{nV_{g,\min}}{I_{o,\max}} = 55.1\ \Omega. \tag{57}$$

6) From (29), we obtain the following expression for the resonant frequency:

$$f_{o1} = \frac{\gamma_1}{\pi} f_s = 158.9 \text{ kHz}. \tag{58}$$

7) From (31), L_s is given by

$$L_s = \frac{Z_{o1}}{\omega_{o1}} = 5.52 \text{ mH} \tag{59}$$

hence, $L_p = 11.04$ mH.

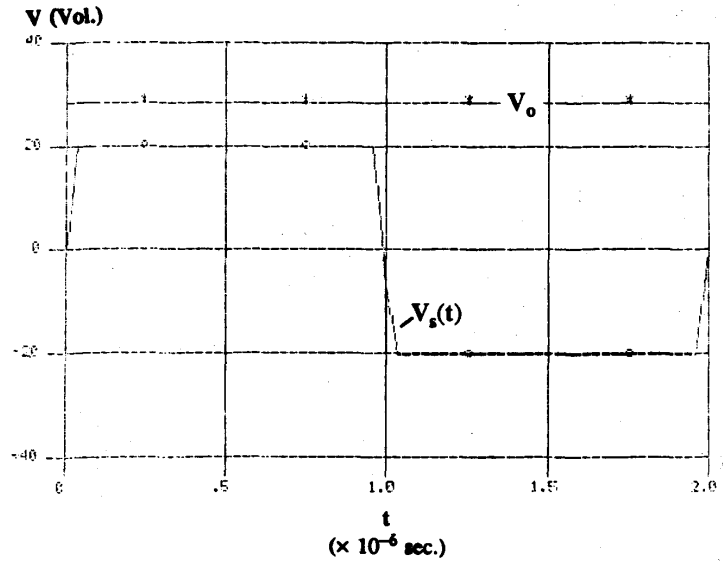
8) From (20), C_s is given by

$$C_s = \frac{L_s}{Z_{o1}^2}\left(1 + \frac{C_s}{C_p} K_1\right) = 6.8\ \mu\text{F} \tag{60}$$

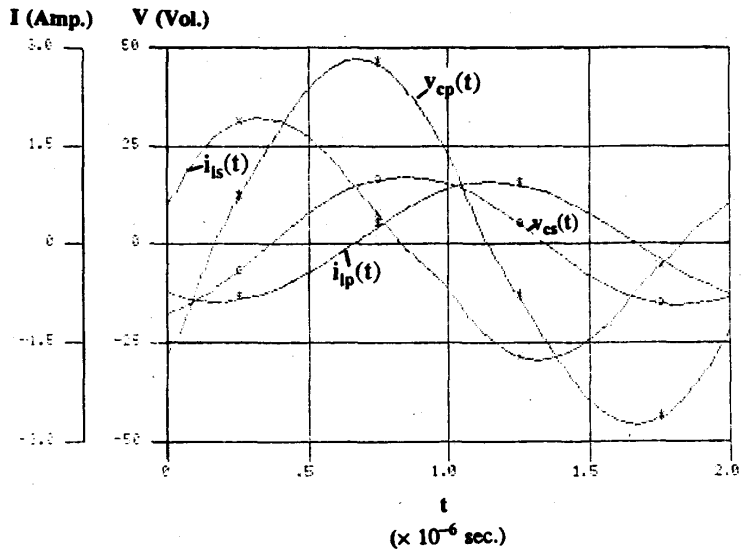
hence, $C_p = 3.4\ \mu$F.

Conclusion

In this paper, a new converter topology known as LLCC-type parallel resonant converter (PRC-LLCC), in which the tank circuit consists of two inductors and two capacitors, was introduced. Using the state-plane approach, we presented the steady state analysis of the PRC-LLCC operating in the continuous conduction mode. It was shown that by using the state variable transformation technique, the steady state response of the converter can be represented by two state-plane diagrams. Using these diagrams, together with the circuit equations, we derived a set of control characteristic curves which are very useful for converter design. Based on these curves, a design pro-



(a)



(b)

Fig. 9. Voltage and current waveforms from computer simulations. (a) $V_s(t)$ and $V_o(t)$. (b) $v_{cp}(t)$, $v_{cs}(t)$, $i_{Ls}(t)$, and $i_{Lp}(t)$.

cedure along with a specific design example have been given. The correctness of our analysis results was verified via computer simulations.

References

[1] I. Batarseh, R. Liu, C. Q. Lee, and Upadhaya, "150 Watts and 140 kHz multi-output LCC-type parallel resonant converter," in *Conf. Rec. IEEE-APEC'89 Annual Meeting*, Baltimore, MD, Mar. 1989, pp. 221–230.

[2] I. Batarseh, "Analysis and design of high order parallel resonant converters," Ph.D. dissertation, University of Illinois at Chicago, Apr. 1990.

[3] A. S. Bhat and S. B. Dewan, "Analysis and design of a high frequency resonant converter using LCC type commutation," in *Conf. Rec. IEEE-APEC*, 1986, pp. 657–663.

[4] I. E. Batarseh, R. Liu, and C. Q. Lee, "State-plane analysis and design of parallel resonant converter with LCC-type commutation," in *Conf. Rec. 27th SICE Annual Meeting*, Narashino, Japan, Aug. 2–4, 1988, pp. 831–835.

[5] R. L. Steigerwald, "A comparison of half-bridge resonant converter topologies," *IEEE Ind. Applications Society, Annual Meeting*, 1978, pp. 135–144.

[6] S. D. Johnson, A. F. Witulski, and R. W. Erickson, "Comparison of resonant topologies in high-voltage DC applications," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 24, no. 3, pp. 263–273, May 1988.

[7] C. Q. Lee, R. Liu, and I. E. Batarseh, "Analysis and design of parallel

resonant converter with LLC-type commutation," to be published in *IEEE Trans. Aerosp. Electron. Syst.*

[8] A. S. Bhat and S. B. Dewan, "A generalized approach for the steady state analysis of resonant inverter," in *Conf. Rec. IEEE-APEC*, pp. 664-671, 1986.

Issa Batarseh (S'83–M'91) was born in Zerka, Jordan, on January 29, 1961. He received the B.S., M.S. and Ph.D. degrees from the University of Illinois at Chicago in 1983, 1985 and 1990, respectively, all in electrical engineering. His thesis topic was in the analysis and design of high order dc-to-dc parallel resonant converters, which are mostly used for computer and aerospace applications.

After completing his Ph.D., he performed Post-Doctorate work for one year in the area of power electronics at the University of Illinois at Chicago. Currently, Dr. Batarseh is Assistant Professor in the Electrical Engineering Department at the University of Central Florida, Orlando. His current interests are power electronics, digital and analog electronic circuits design, and power semiconductor devices.

C. Q. Lee (M'64) received the B.S., M.S. and Ph.D. degrees from the Illinois Institute of Technology, Chicago, in 1957, 1959, and 1966, respectively, all in electrical engineering.

From 1957 to 1965, he worked at Admiral Corporation, Chicago; Motorola Incorporated, Chicago; and Bell Telephone Laboratories, New Jersey, in the area of waveguide circuitry and microwave diode applications. Since 1965 he has been a member of the faculty of the Department of Electrical Engineering and Computer Science of the University of Illinois at Chicago. Dr. Lee is currently an Associate Professor and the Director of Power Electronic Research Laboratory of the Department where he engages in research and development of high frequency resonant dc-to-dc converters.